

Multimodal Clustering for Disease-Phenotype Discovery

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Data Analysis

Introduction & Motivation

The problem

- discover hidden disease patterns
i.e. phenotype

Why?

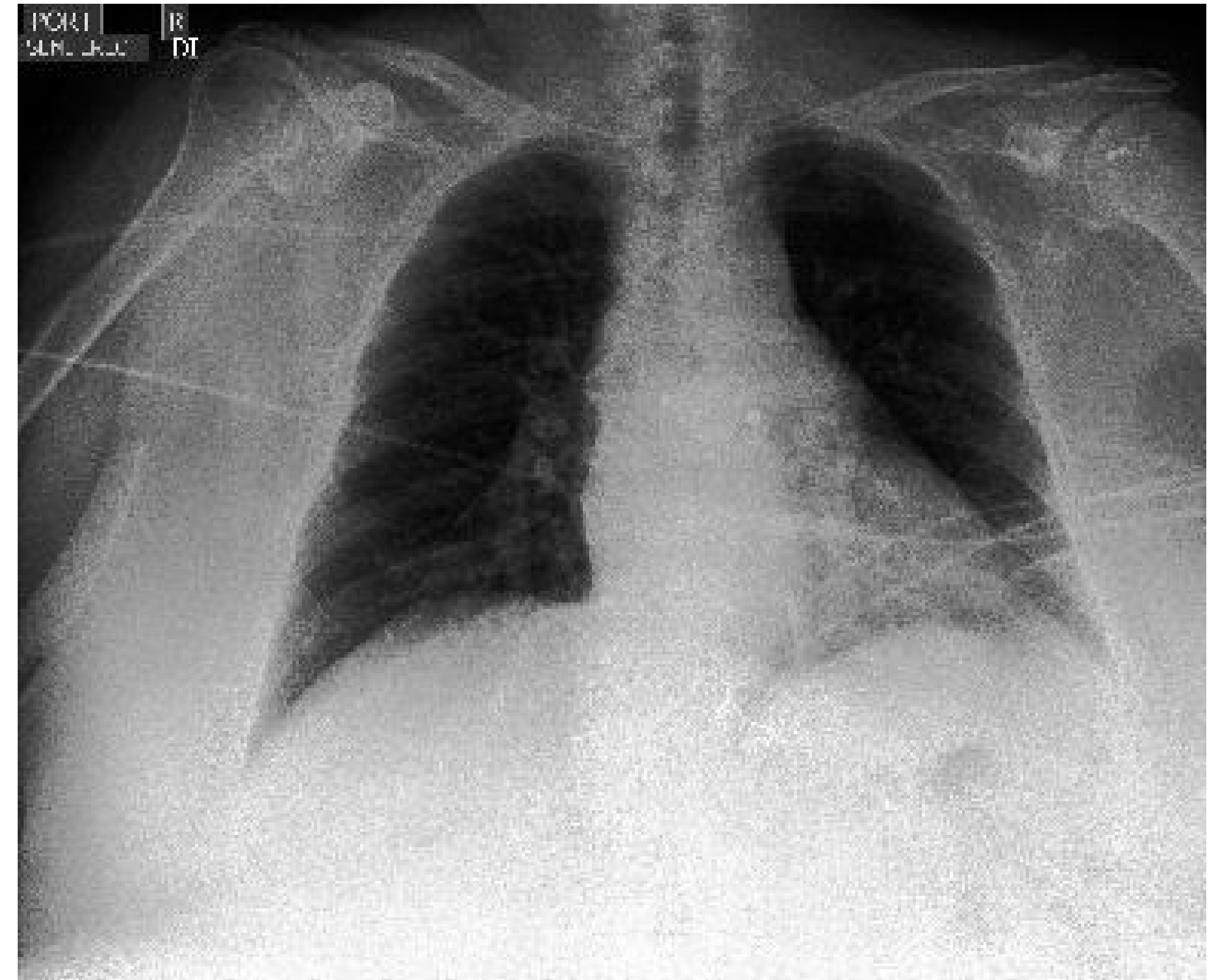
- Medical datasets contain multiple data types
- Traditional approaches often analyze these separately
- Patients with the same diagnosis may still have different visual and clinical patterns.

Relevance

- Diseases of the respiratory system accounted for 7.0% of all deaths in the EU in 2022.

Dataset

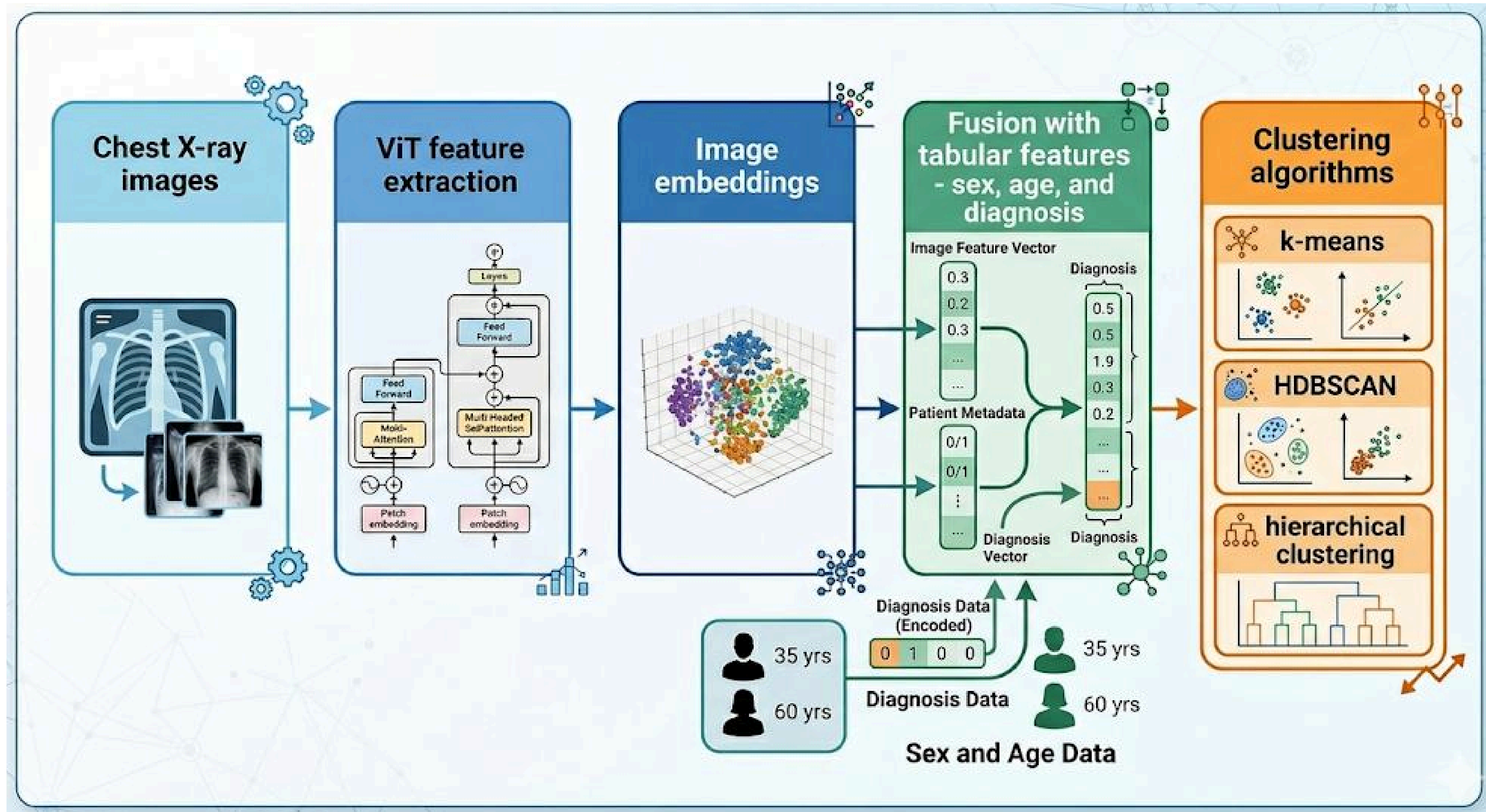
CheXpert & NIH Chest X-rays



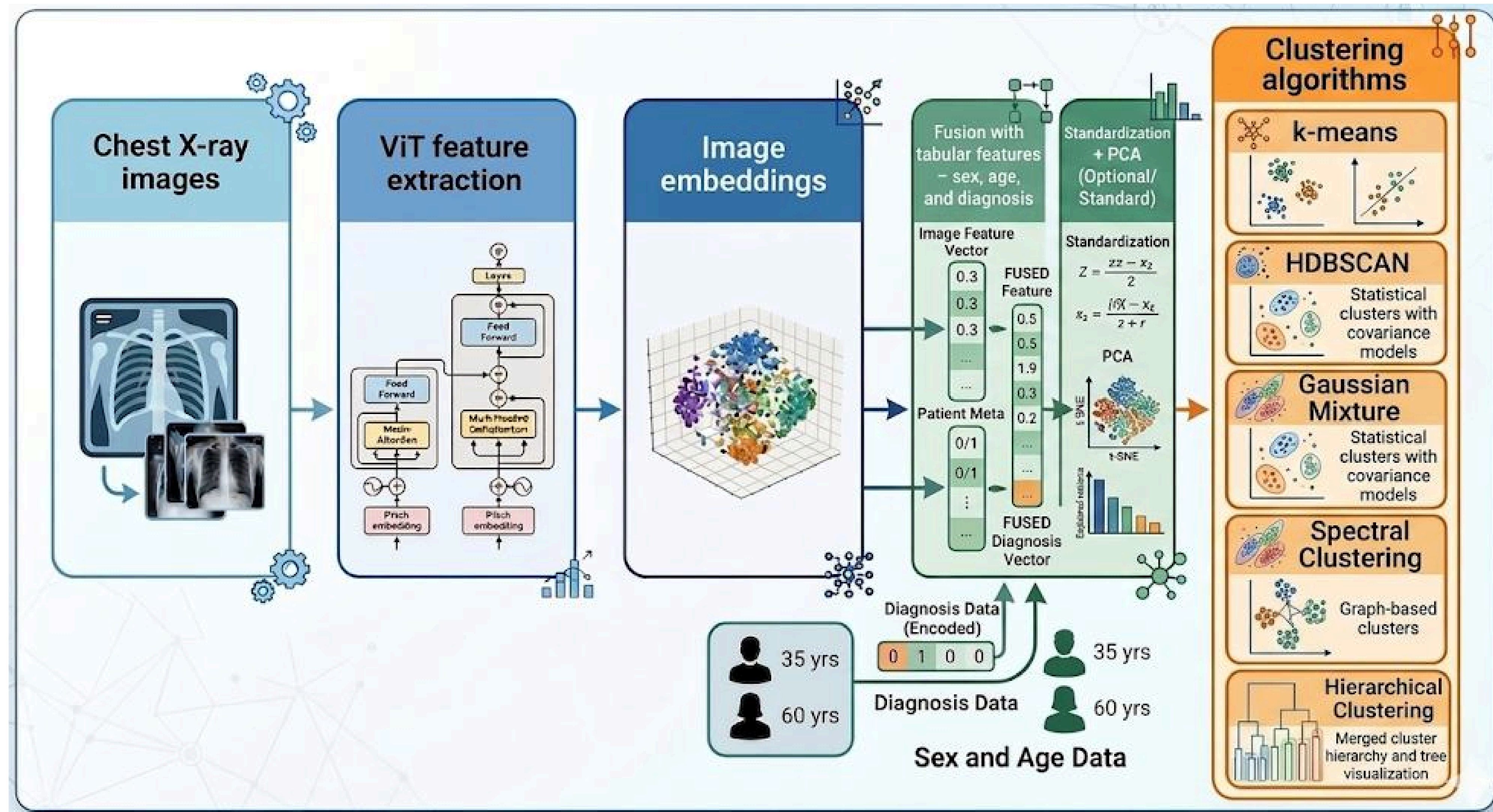
Composition

- images
- 12 diagnosis: **Pneumonia, Edema, Cardiomegaly** etc.
- tabular: **sex, age**

Methodology



Improvements

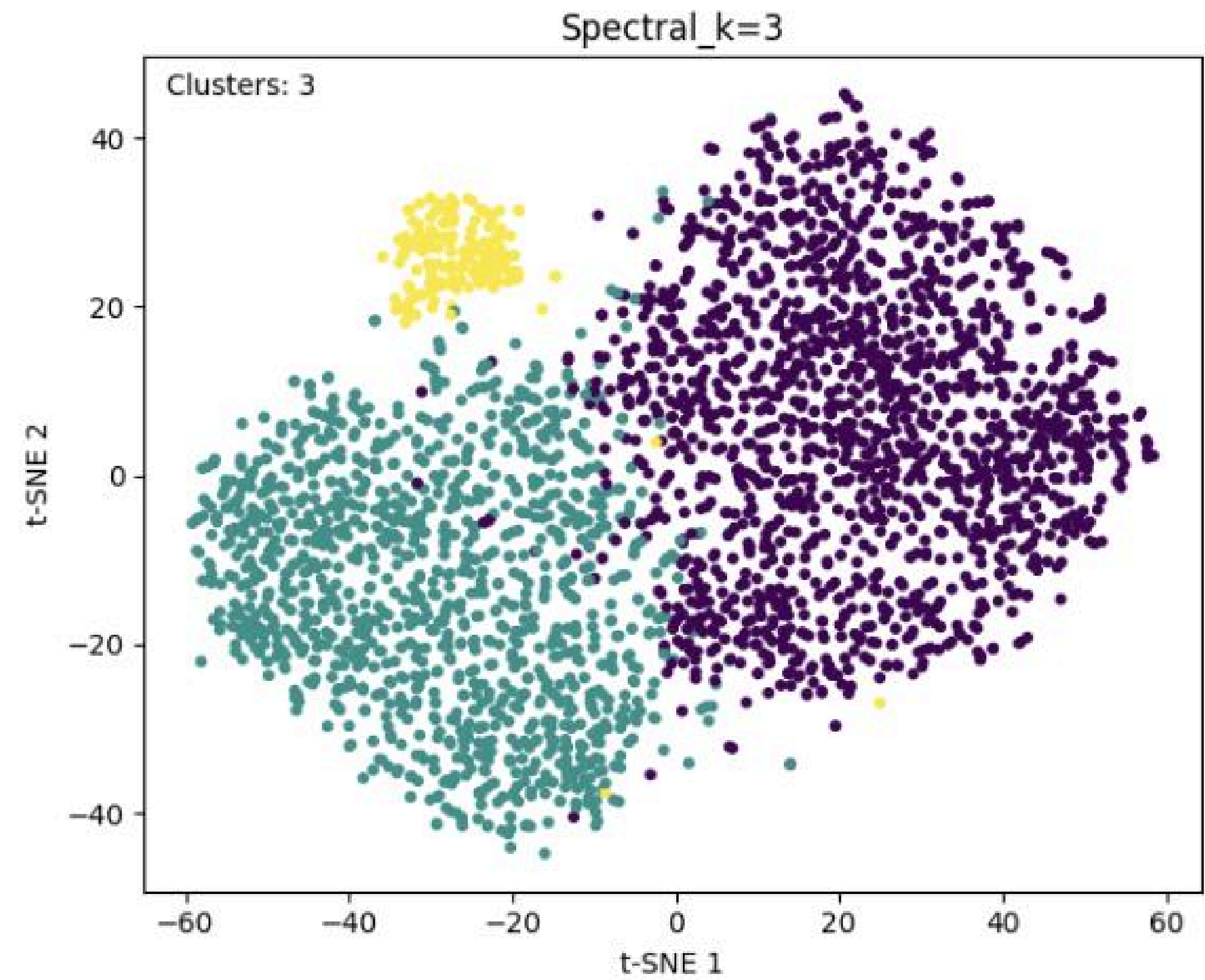
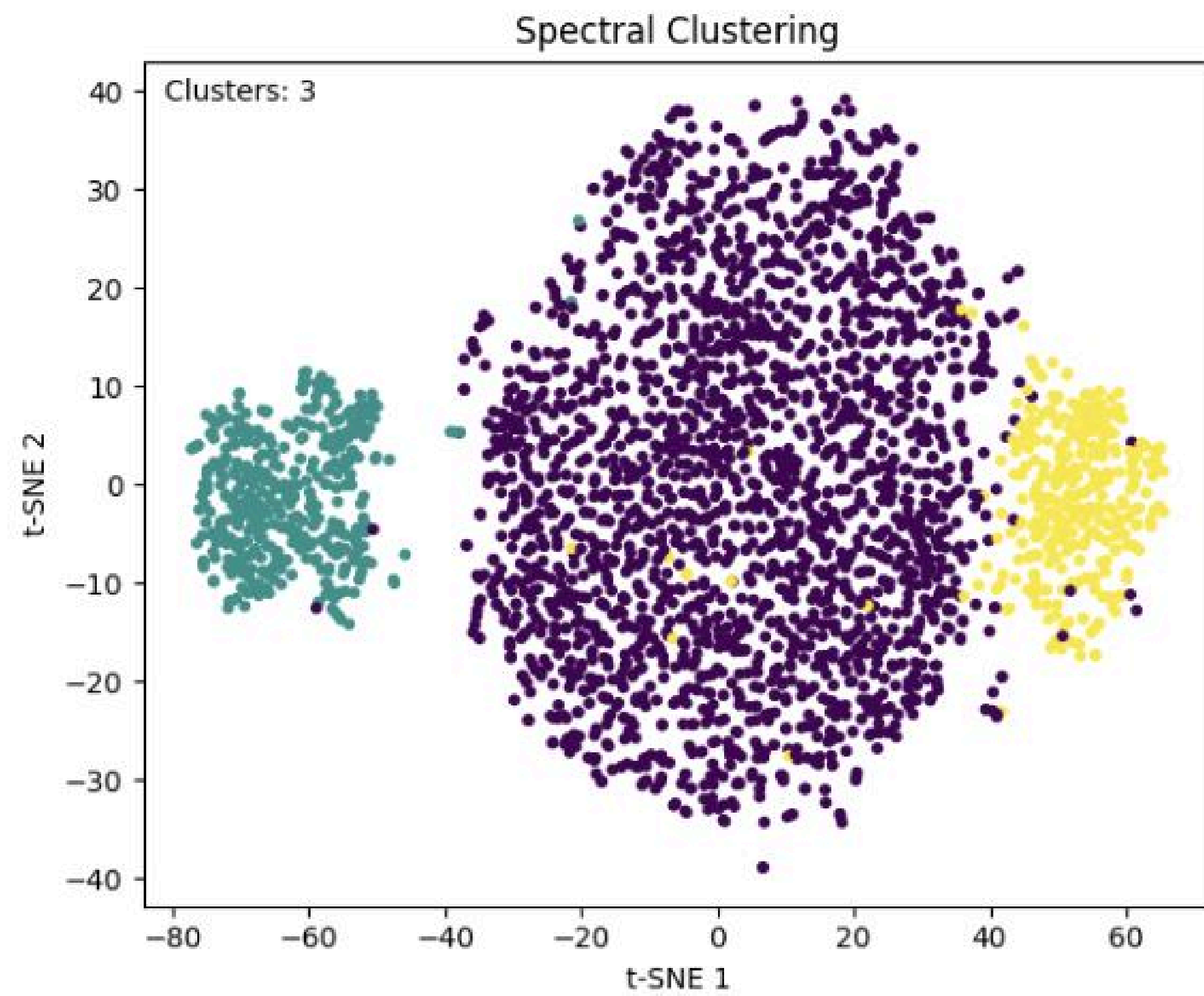


Results

Algorithm	n_clusters	noise_points	Silhouette	Davies-Bouldin	Calinski-Harabasz
HDBSCAN	2	1590	0.179	2.195	1073.59
HC	3	0	0.053	3.431	1247.296
KMeans	3	0	0.072	3.047	1434.13

Algorithm	Clusters	Noise	Silhouette	DBI	CHI
HDBSCAN	2	1456	0.209	1.888	1577.55
HC	3	0	0.102	2.520	1556.95
K-Means	3	0	0.098	2.605	1950.97
Spectral Clustering	3	0	0.096	2.335	1533.18
GMM	3	0	0.066	3.131	1591.96

Results



Interpretation

Cluster	Disease	Cluster Size	Disease Count	Cluster Prev.	Lift
0	Fibrosis	4580	244	0.053	1.483
0	Hernia	4580	28	0.006	1.423
0	Pleural Thickening	4580	372	0.081	1.255
0	Nodule	4580	672	0.147	1.164
0	Pneumothorax	4580	384	0.084	1.148
1	Edema	3367	262	0.078	2.063
1	Consolidation	3367	430	0.128	1.544
1	Atelectasis	3367	944	0.280	1.202
1	Effusion	3367	934	0.277	1.137
1	Pneumonia	3367	93	0.028	1.123
2	Hernia	430	4	0.009	2.165
2	Pneumothorax	430	47	0.109	1.496
2	Emphysema	430	24	0.056	1.347
2	Cardiomegaly	430	29	0.067	1.329
2	Pleural Thickening	430	35	0.081	1.258

Impact

Industry applications

- hospital analytics
- clinical research
- radiology decision-support tools

Value

- discover hidden patient subgroups;
- identify rare or unusual disease patterns;
- improve understanding of disease heterogeneity;
- help hospitals use existing imaging data more effectively.

Thank you!